



Department of Computer Science and Engineering
Lab Report

Lab Report No : 7				
Lab Report Name : Unsupervised learning algorithms (K-Means, Hierarchical Clustering, and DBSCAN) using customer segmentation dataset				
Course Title : Artificial Intelligence Sessional				
Course Code : CSE 422				
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Level : 4		Term : 2		Section : A
				Group : G1
Date of Submission : 19.07.2026		Semester : Spring		Year : 2026

Marking Rubric

Problem Understanding & Report Clarity (3)	Implementation (4)	Results, Analysis & Presentation (3)	Total (10)

Key Learnings:

- I learned how to implement and compare K-Means, Hierarchical Clustering, and DBSCAN for customer segmentation.
- I learned how to use the Elbow Method to determine the optimal number of clusters for K-Means.
- I learned how to evaluate clustering performance using the Silhouette Score and Davies–Bouldin Index.
- I learned that K-Means provided the best clustering performance for the Mall Customer dataset.

Code Implementation:

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STEP 1: DATASET OVERVIEW
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Shape: (200, 5)

First 5 rows:
  CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)
0           1    Male   19                15                39
1           2    Male   21                15                81
2           3  Female   20                 16                 6
3           4  Female   23                 16                77
4           5  Female   31                 17                40

Data Types:
CustomerID      int64
Gender          object
Age             int64
Annual Income (k$)  int64
Spending Score (1-100)  int64
dtype: object

Missing Values:
CustomerID      0
Gender          0
Age             0
Annual Income (k$)  0
Spending Score (1-100)  0
dtype: int64

Statistical Summary:
   CustomerID      Age  Annual Income (k$)  Spending Score (1-100)
count  200.000000  200.000000    200.000000    200.000000
mean    100.500000   38.850000     60.560000     50.200000
std     57.879185   13.969007     26.264721     25.823522
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```

Figure No: 01

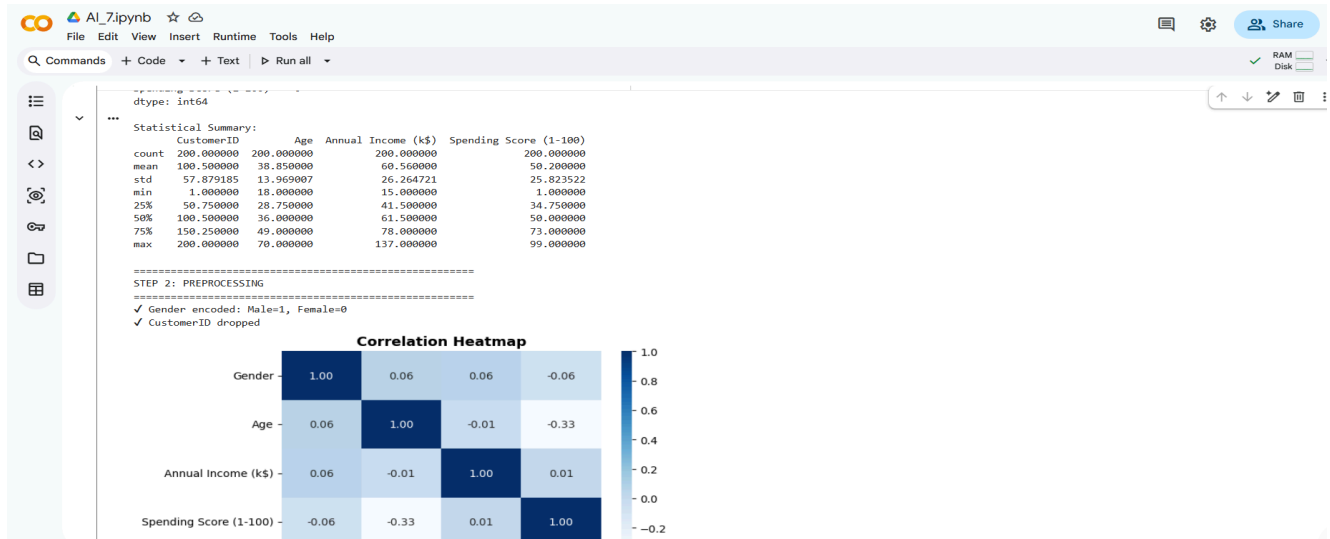


Figure No: 02

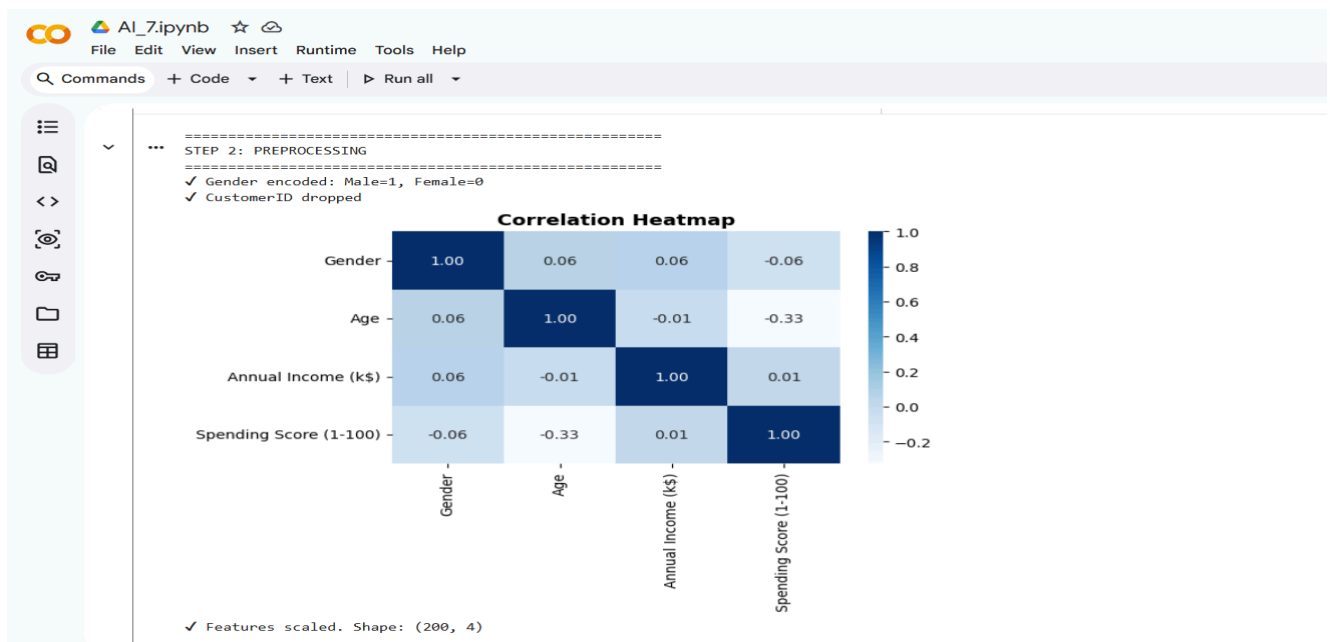


Figure No: 03

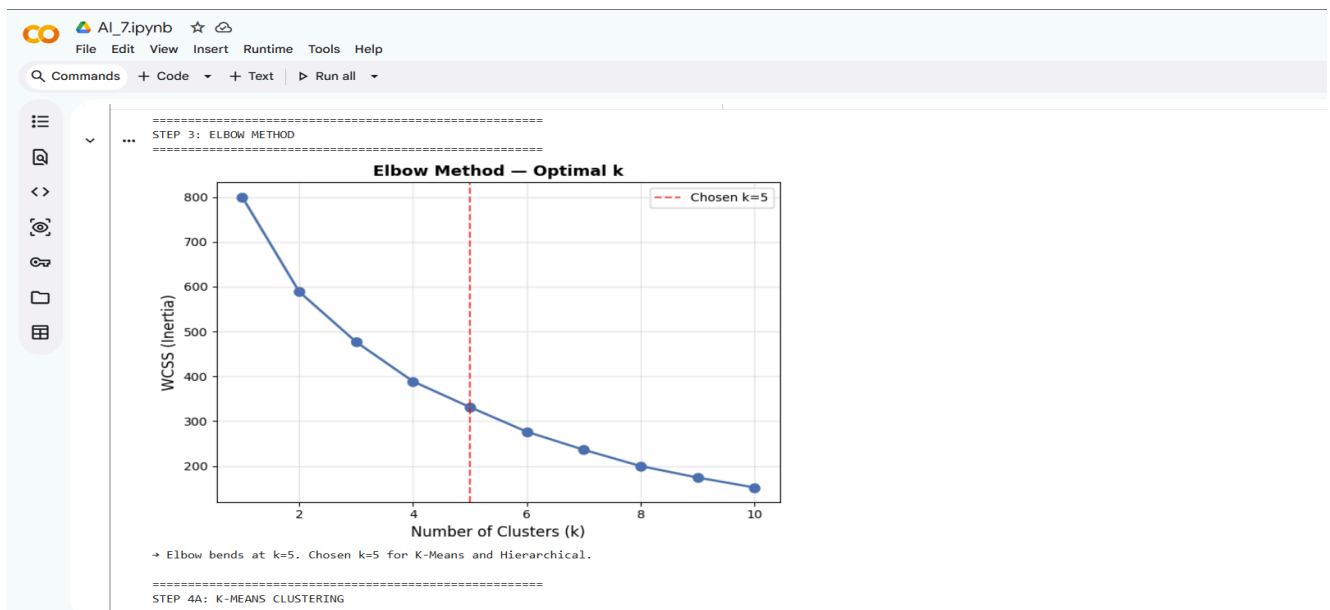


Figure No: 04



Figure No: 05

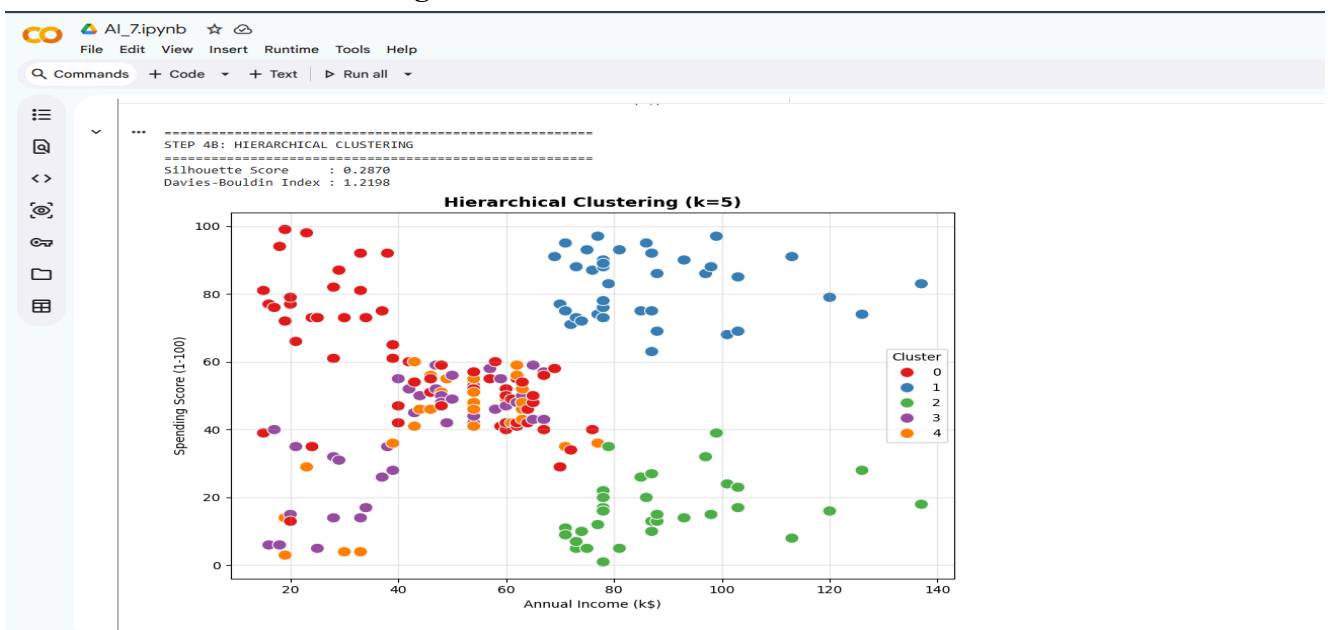


Figure No: 06

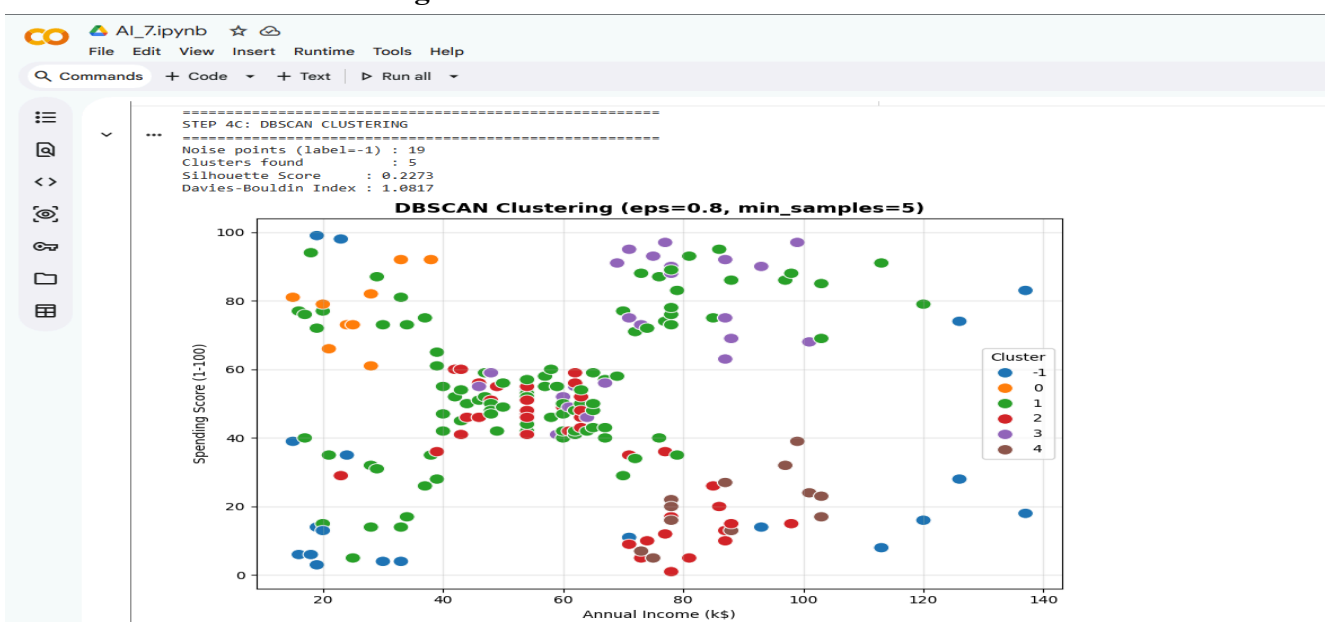


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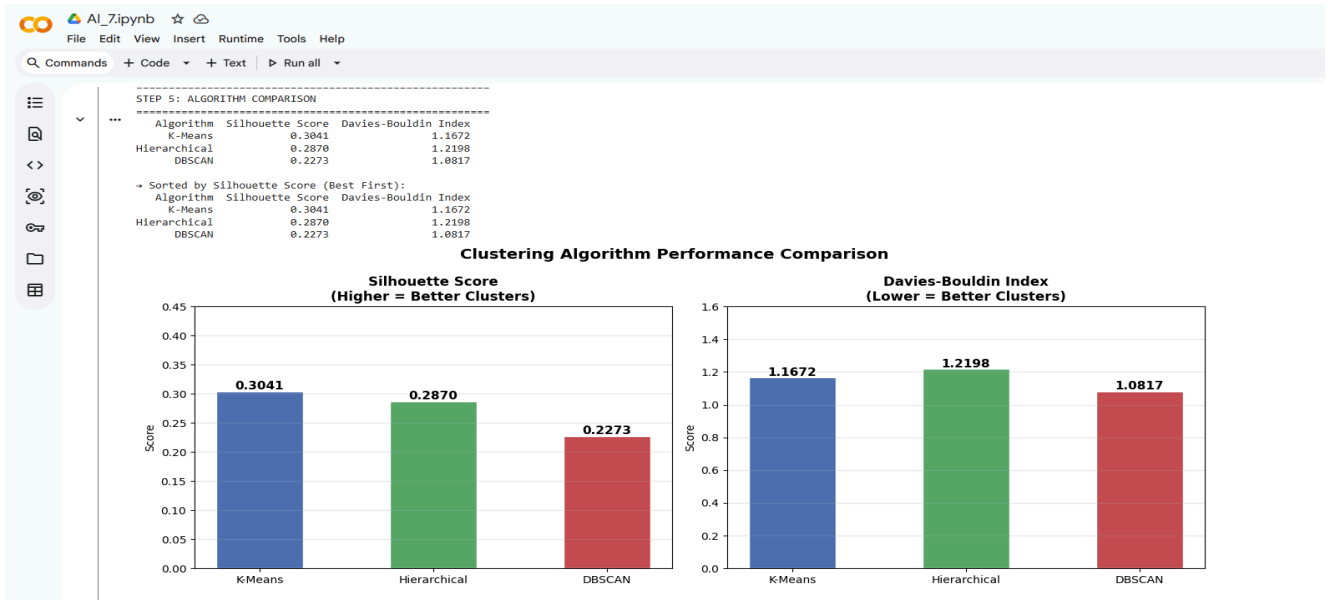


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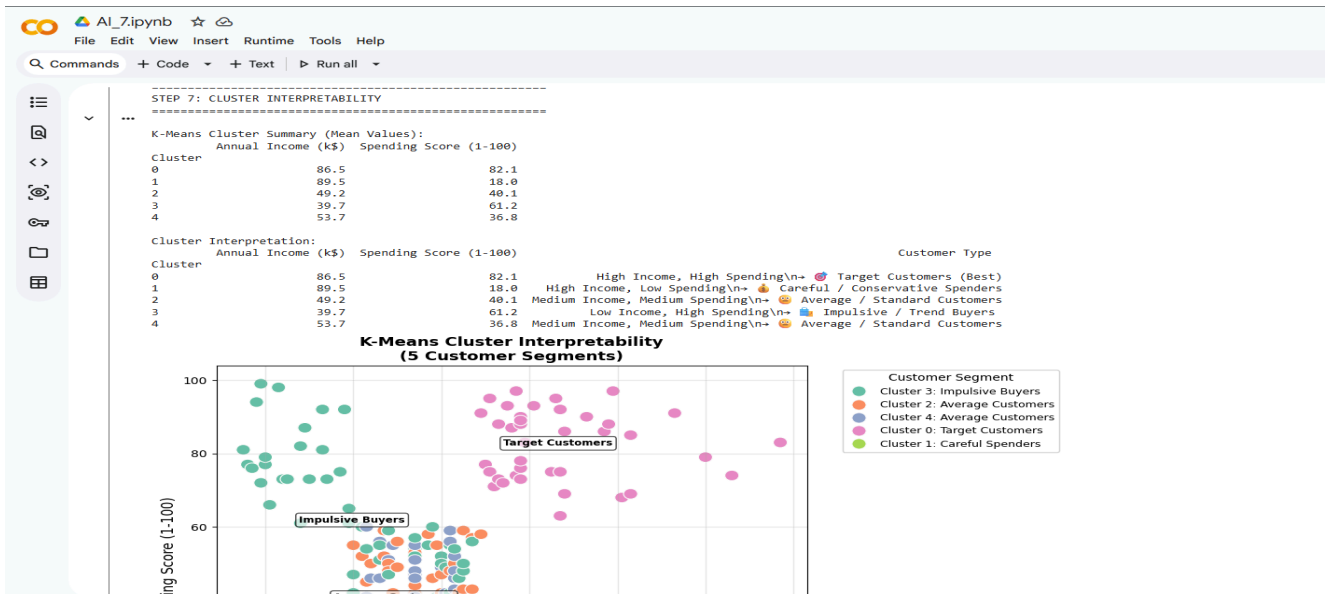


Figure No: 09

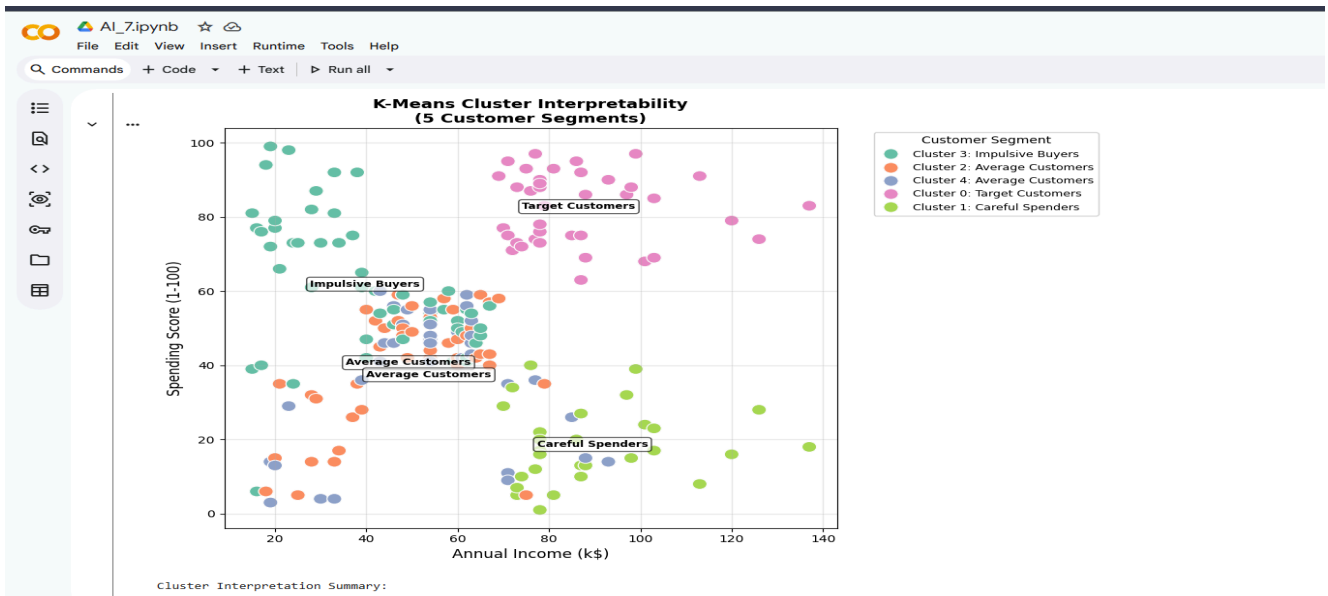


Figure No: 10

Cluster Interpretation Summary:

- Target Customers—High Income + High Spending
→ Best marketing priority. Offer premium deals.
- Careful Spenders—High Income + Low Spending
→ Potential high-value customers. Need incentives.
- Impulsive Buyers—Low Income + High Spending
→ Respond well to flash sales & trendy products.
- Budget Customers—Low Income + Low Spending
→ Price-sensitive. Attract with discounts.
- Average Customers—Medium Income + Medium Spending
→ General audience. Standard promotions work.

Result Analysis

- K-Means produced the BEST Silhouette Score (0.3041) and the second-lowest Davies-Bouldin Index (1.1672).
 - Clear, compact, well-separated clusters.
- Hierarchical clustering performed slightly lower than K-means on both metrics. Still reasonable but not optimal here.
- DBSCAN labeled 19 points as noise and scored the lowest silhouette score. This dataset has compact spherical clusters.
 - Not suited for density-based separation.
- Cluster interpretability confirmed 5 meaningful business segments: Target Customers, Careful Spenders, Impulsive Buyers, Budget Customers, and Average Customers.
- So that we can say, K-Means is the best algorithm for mall customer segmentation based on metrics, visual separation, and cluster interpretability.